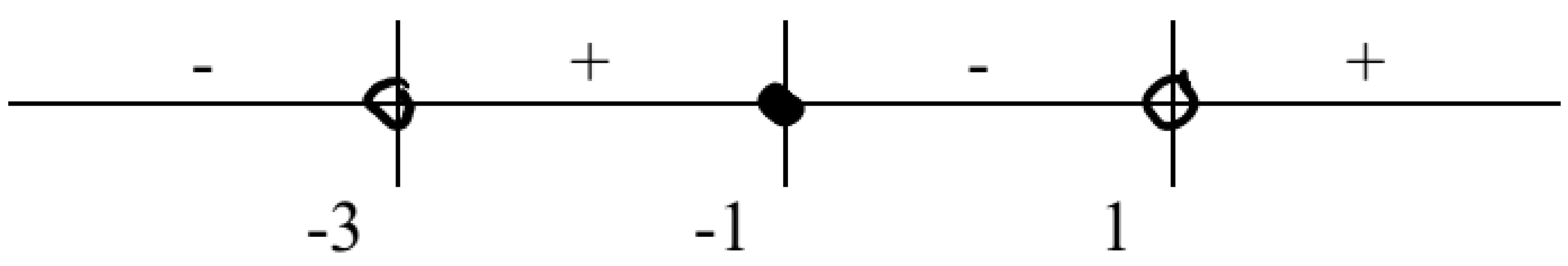
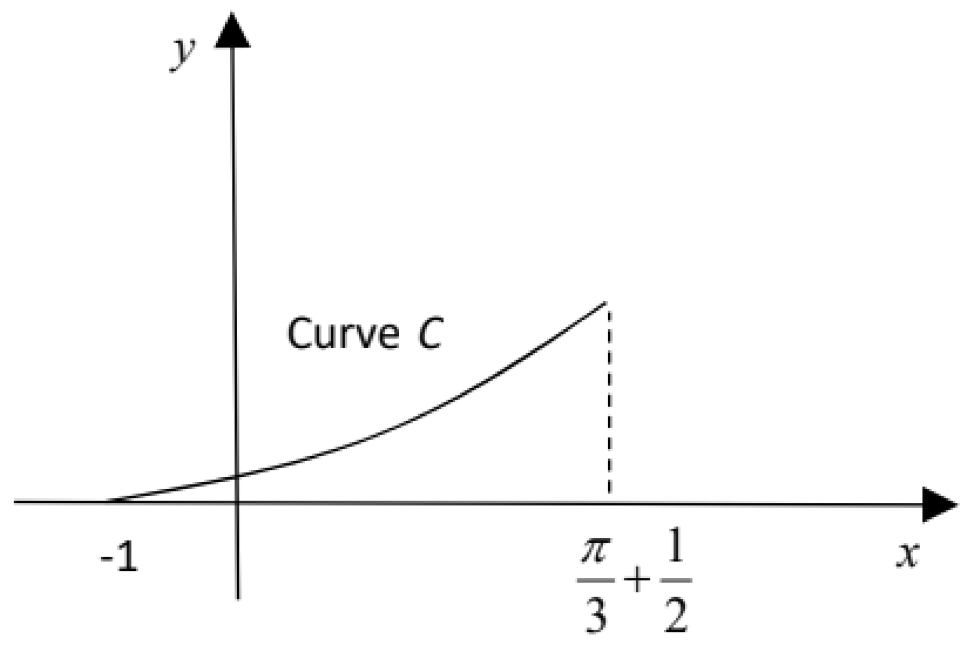
Hwa Chong Institution (College)

2017 C1 H2 Maths Promo Exam Solution

Qtn	Solution	Remarks
1(i)	$\dfrac{x+2}{x-1} \le \dfrac{x}{x+3}$ $\dfrac{(x+2)(x+3)-x(x-1)}{(x-1)(x+3)} \le 0$ $\dfrac{x^2+5x+6-x^2+x}{(x-1)(x+3)} \le 0$ $\dfrac{6(x+1)}{(x-1)(x+3)} \le 0$  $x<-3$ or $-1 \le x < 1$	This question was generally well done. There were some who were careless in the combining of the 2 fractions or who erroneously perform cross multiplication for the inequality without due thought for the signs of the terms. Quite a number of students did not omit the values $x=-3$ and $x=1$ in their answers. Students are to note also that the word used is 'or' in the answer '$x<-3$ **OR** $-1 \le x < 1$' and that it would be wrong to write '$x<-3$ and $-1 \le x < 1$'.
(ii)	$\dfrac{e^x+2}{e^x-1} \le \dfrac{e^x}{e^x+3}$ Replace x with e^x, $e^x < -3$ (N.A.) or $-1 \le e^x < 1$ $\Rightarrow e^x < 1$ $\Rightarrow x < 0$ $\therefore \{x \in \mathbb{R} : x < 0\}$	There was an unnecessary loss of marks as students left their answer as $x<0$ and forgot to write the final answer as $\{x \in \mathbb{R} : x < 0\}$ in SET notation as required by question. Students need to recognize that $e^x < -3 \Rightarrow$ no solution as $e^x > 0$ for all real x $e^x \ge -1 \Rightarrow x \in \mathbb{R}$ as $e^x > 0$ for all real x Do note that $-1 \le e^x < 1 \Rightarrow e^x < 1$ as the lower bound '-1' does not restrict values of x as $e^x > 0$ for all real x. Some students left their answer as $\{x \in \mathbb{R} : x < \ln 1\}$ when they do need to write ln 1 as 0.

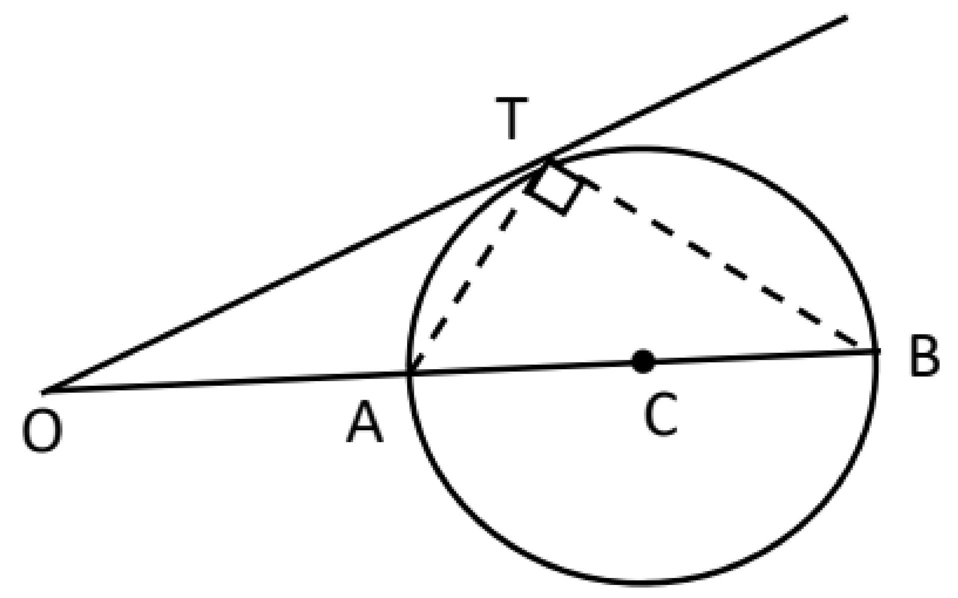
Hwa Chong Institution (College)

2	 $x=t-\cos 2t$ $\frac{\mathrm{d}x}{\mathrm{d}t}=1+2\sin 2t$ Area $=\int_{-1}^{\frac{\pi}{3}+\frac{1}{2}} y\,\mathrm{d}x$ $=\int_0^{\frac{\pi}{3}} \sin^2 t(1+2\sin 2t)\,\mathrm{d}t$ $=\int_0^{\frac{\pi}{3}} \left(\sin^2 t+2\sin^2 t\sin 2t\right)\mathrm{d}t$ (Shown)	A worrying number of students attempted in vain to convert the given parametric equations to cartesian equation while some were totally clueless. Students should first write down the area as $\int_{-1}^{\frac{\pi}{3}+\frac{1}{2}} y\,\mathrm{d}x$ which then becomes $\int_0^{\frac{\pi}{3}} \sin^2 t(1+2\sin 2t)\,\mathrm{d}t$ after changing to variable t. Many students did not change the limits for x in $\int_{-1}^{\frac{\pi}{3}+\frac{1}{2}} y\,\mathrm{d}x$ to limits for t in $\int_0^{\frac{\pi}{3}} \sin^2 t(1+2\sin 2t)\,\mathrm{d}t$.
	$\int_0^{\frac{\pi}{3}} \left(\sin^2 t+2\sin^2 t\sin 2t\right)\mathrm{d}t$ $=\int_0^{\frac{\pi}{3}} \sin^2 t\,\mathrm{d}t+2\int_0^{\frac{\pi}{3}} \sin^2 t\sin 2t\,\mathrm{d}t$ $=\int_0^{\frac{\pi}{3}} \frac{1-\cos 2t}{2}\,\mathrm{d}t+2\int_0^{\frac{\pi}{3}} \sin^2 t(2\cos t\sin t)\,\mathrm{d}t$ $=\left[\frac{t}{2}-\frac{\sin 2t}{4}\right]_0^{\frac{\pi}{3}}+4\int_0^{\frac{\pi}{3}} \cos t(\sin^3 t)\,\mathrm{d}t$ $=\frac{\pi}{6}-\frac{1}{4}\sin\frac{2\pi}{3}+4\left[\frac{\sin^4 t}{4}\right]_0^{\frac{\pi}{3}}$ $=\frac{\pi}{6}-\frac{1}{4}\left(\frac{\sqrt{3}}{2}\right)+\left(\frac{\sqrt{3}}{2}\right)^4$ $=\frac{\pi}{6}+\frac{9-2\sqrt{3}}{16}$	For $\int_0^{\frac{\pi}{3}} \sin^2 t\,\mathrm{d}t$, we can convert $\sin^2 t$ to $\frac{1-\cos 2t}{2}$ using MF26. Some students showed working such as $\sin^2 t=1-\cos^2 t=1-\frac{1+\cos 2t}{2}=\frac{1-\cos 2t}{2}$ which is unnecessarily lengthy. Students who solved using 'by parts' displayed long and laborious working which is rather inefficient for this question. It was a pity when students made error in writing $\frac{\pi}{6}-\frac{\sqrt{3}}{8}+\frac{9}{16}$ as $\frac{\pi}{6}-\frac{2\sqrt{3}+9}{16}$.
3(a)	**Method 1** $\frac{3}{2}\int_0^b \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2-x^2}}\,\mathrm{d}x=\frac{\pi}{2}$	Students have either forgotten to take square root of 4 (for method 1) or divide by 2 (for method 2).

Hwa Chong Institution (College)

	$\left[\frac{3}{2}\sin^{-1}\left(\frac{x}{\frac{1}{2}}\right)\right]_0^b = \frac{1}{2}\pi$ $\left[\sin^{-1}2x\right]_0^b = \frac{1}{3}\pi$ $\sin^{-1}2b = \frac{1}{3}\pi$ $2b = \sin\frac{\pi}{3}$ $b = \frac{\sqrt{3}}{4}$ **Method 2** $\frac{3}{2}\int_0^b \frac{2}{\sqrt{1-(2x)^2}}\,\mathrm{d}x = \frac{\pi}{2}$ $\left[\sin^{-1}2x\right]_0^b = \frac{1}{3}\pi$ $\sin^{-1}2b = \frac{1}{3}\pi$ $2b = \sin\frac{\pi}{3}$ $b = \frac{\sqrt{3}}{4}$	
3(b)	$\int \frac{1}{\sqrt{4+x^2}}\,\mathrm{d}x$ Let $x = 2\tan\theta \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\sec^2\theta$ $\int \frac{1}{\sqrt{4+x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{4+4\tan^2\theta}}\left(2\sec^2\theta\right)\mathrm{d}\theta$ $= \int \frac{1}{2\sec\theta}\left(2\sec^2\theta\right)\mathrm{d}\theta$ $= \int \sec\theta\,\mathrm{d}\theta$ $= \ln\left\|\sec\theta + \tan\theta\right\| + C$ $= \ln\left\|\sqrt{1+\frac{x^2}{4}} + \frac{x}{2}\right\| + C$	Some students have forgotten the square root in the denominator in the process of simplifying. Many students did not give final answer in terms of x. A significant of students tried to express $\sec\theta$ in terms of x by writing it as $\sec\left(\tan^{-1}\frac{x}{2}\right)$ which is unaccepted. Some students thought that $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ is the lower and upper limit of the integral. Still a significant number of students forgot $+ C$ in their final answer.

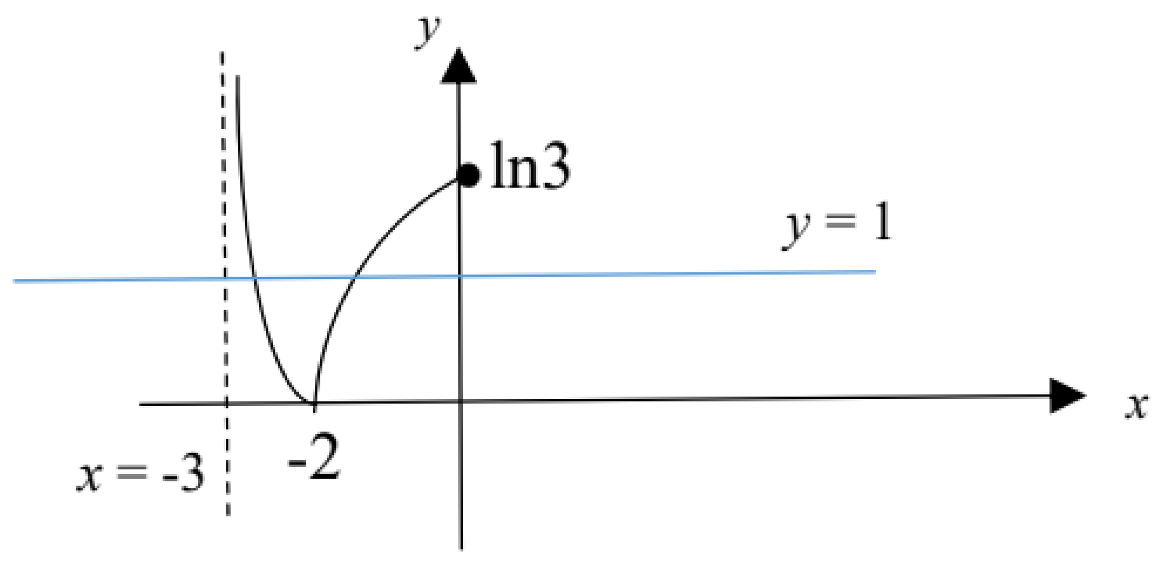
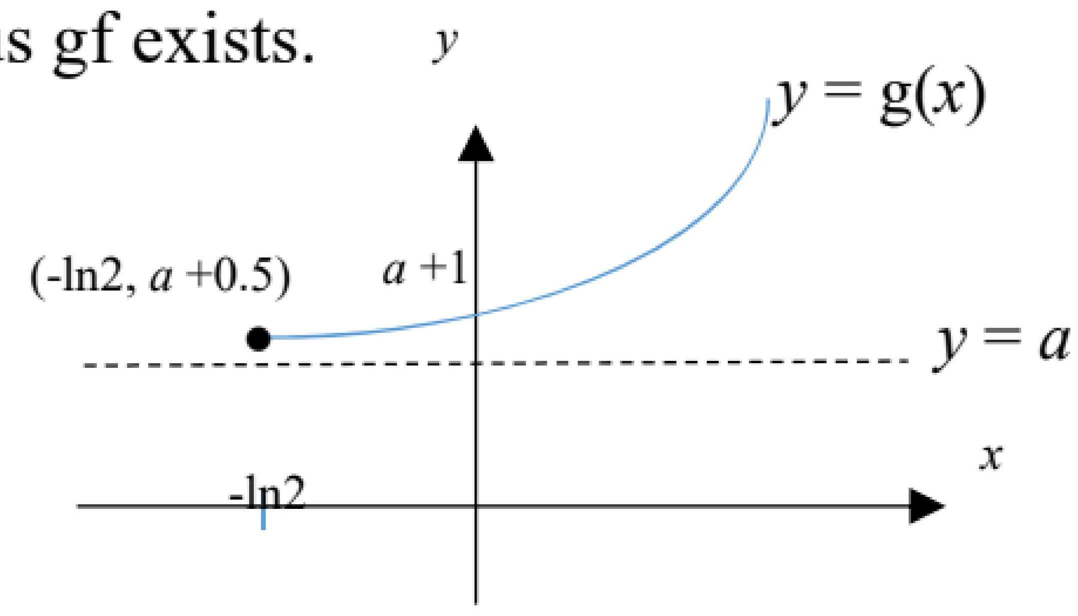
Hwa Chong Institution (College)

4(i)	 AT is perpendicular to TB. $(\mathbf{t}-\mathbf{a})\bullet(\mathbf{t}-\mathbf{b})=0$ $(\mathbf{t}-\mathbf{a})\bullet(\mathbf{t}-k\mathbf{a})=0$ (Since $\mathbf{b}=k\mathbf{a}$) $\|\mathbf{t}\|^2-\mathbf{a}\bullet\mathbf{t}-k\mathbf{a}\bullet\mathbf{t}+k\|\mathbf{a}\|^2=0$ $\mathbf{a}\bullet\mathbf{t}=\dfrac{k\|\mathbf{a}\|^2+\|\mathbf{t}\|^2}{k+1}$	Quite well done. Students need to be more careful with their presentation. Many students did not put '.' for dot product.
(ii)	$\measuredangle OTC$ is 90°. $\Rightarrow \overrightarrow{OT}\bullet\overrightarrow{TC}=0$ $\mathbf{t}\bullet(\mathbf{c}-\mathbf{t})=0$ Using Ratio Theorem, $\overrightarrow{OC}=\dfrac{\overrightarrow{OA}+\overrightarrow{OB}}{2}=\dfrac{\mathbf{a}+k\mathbf{a}}{2}=\dfrac{(k+1)\mathbf{a}}{2}$ $\mathbf{t}\bullet\left(\dfrac{(k+1)\mathbf{a}}{2}-\mathbf{t}\right)=0$ $\dfrac{(k+1)}{2}\mathbf{a}\bullet\mathbf{t}=\|\mathbf{t}\|^2$ $\dfrac{(k+1)}{2}\left(\dfrac{k\|\mathbf{a}\|^2+\|\mathbf{t}\|^2}{k+1}\right)=\|\mathbf{t}\|^2$ $k\|\mathbf{a}\|^2+\|\mathbf{t}\|^2=2\|\mathbf{t}\|^2$ $k\|\mathbf{a}\|^2=\|\mathbf{t}\|^2$ $\dfrac{\|\mathbf{a}\|}{\|\mathbf{t}\|}=\dfrac{1}{\sqrt{k}}\Rightarrow\|\mathbf{a}\|:\|\mathbf{t}\|=1:\sqrt{k}$	Some students thought that $\overrightarrow{OC}=\dfrac{1}{2}\overrightarrow{AB}$. Many students do not understand the meaning of ratio, hence give the wrong final answer. There are still students who write $\mathbf{t}\bullet\mathbf{t}=\mathbf{t}^2$ instead of $\mathbf{t}\bullet\mathbf{t}=\|\mathbf{t}\|^2$.

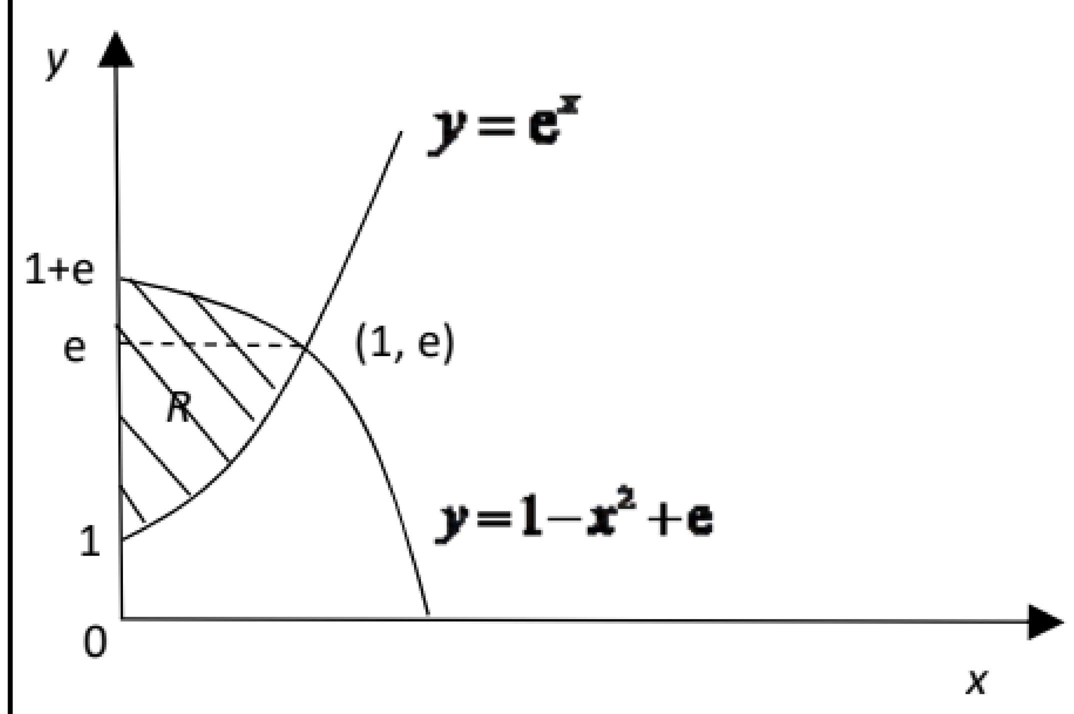
Hwa Chong Institution (College)

5(i)	$u_1=3,\ u_2=9,\ u_3=18.$	Some students wrote down the pattern but did not write the final solution for U_2 and U_3, and hence were not awarded the credit.
(ii)	Given that $u_{r+1}=u_r+ar+b$ and $u_1=3,\ u_2=9,\ u_3=18$, we have $u_2=u_1+a+b$ and $u_3=u_2+2a+b$. $\Rightarrow 9=3+a+b$ and $18=9+2a+b$ Solving the equations, $a=3,\ b=3$. (Shown)	A handful of students wrote $U_{r+1}=U_r+3r+3$ without any proof. Though it is obvious but the examiner may just take it that students just took the answer to plug into this relationship. Other students tried to substitute $U_0=0$ into the relationship. Take note that the relationship is only valid for $r\in\mathbb{Z}^+$, and hence U_0 is undefined.
(iii)	$\sum_{r=1}^{n-1}(3r+3)=6+9+12+...+(3(n-1)+3)$ $=\frac{n-1}{2}[6+3n]$ $=\frac{3(n-1)(n+2)}{2}$ $u_{r+1}-u_r=3r+3$ $\Rightarrow\sum_{r=1}^{n-1}(u_{r+1}-u_r)=\sum_{r=1}^{n-1}(3r+3)$ $\sum_{r=1}^{n-1}(3r+3)=\sum_{r=1}^{n-1}(u_{r+1}-u_r)$ $=\cancel{u_2}-u_1$ $+\cancel{u_3}-\cancel{u_2}$ $+u_4-\cancel{u_3}$ $+\cancel{...}$ $+u_n-\cancel{u_{n-1}}$ $=u_n-u_1$ $u_n-u_1=\frac{3(n-1)(n+2)}{2}$ ---------(*) $u_n=\frac{3(n-1)(n+2)}{2}+u_1$ $=\frac{3(n-1)(n+2)}{2}+3$ $=\frac{3(n^2+n-2+2)}{2}$ $=\frac{3n(n+1)}{2}$	Most students are able to get the first part correct. A handful of students were able to identify that $\sum_{r=1}^{n-1}(3r+3)=\sum_{r=1}^{n-1}(u_{r+1}-u_r)$, and hence used MOD to find $\sum_{r=1}^{n-1}(3r+3)$, which is also awarded full credit. For the next part, the majority of the students were unable to identify the use of MOD in this question. Instead, a large number of students mistook $\sum_{r=1}^{n-1}(3r+3)$ as U_{n-1}, which resulted in an erroneous answer when they substitute it into the relationship in part (ii). A handful of students also mistook $\sum_{r=1}^{n-1}(3r+3)$ as S_{n-1} and used the formula $U_n=S_n-S_{n-1}$, which was wrong. Some students also managed to identify the recurrence relation by showing some working and got $U_n=U_1+\sum_{r=1}^{n-1}(3r+3)$ without the use of MOD. However, those who did not explain how they got the aforementioned will not be given full credit.

Hwa Chong Institution (College)

		Lastly, no marks was awarded for the second part of this question if the students fail to use $\sum_{r=1}^{n-1}(3r+3)$ in their working.
6 (i)	 The horizontal line $y = 1$ that cuts the graph $y = \mathrm{f}(x)$ at more than one point. Thus f is not one-one function, f^{-1} does not exist.	Need to use either (1) horizontal line test or (2) $\mathrm{f}\left(x_1\right)=\mathrm{f}\left(x_2\right)=y$
(ii)	-2	Some students give the range of value of x, instead of giving the maximum value
(iii)	$y=\lvert\ln(x+3)\rvert$ Since $-3<x\le -2, \ln(x+3)<0, \therefore y=-\ln(x+3)$. $\mathrm{e}^{-y}=x+3$ $x=\mathrm{e}^{-y}-3$ $\mathrm{f}^{-1}: x \mapsto \mathrm{e}^{-x}-3,\ \ x\ge 0$	Students need to observe the domain of f to determine the sign for $\ln(x+3)$. For determining the domain of f^{-1}, recall $\mathrm{R_f}=\mathrm{D_{f^{-1}}}$
(iv)	gf exists $\Leftrightarrow R_\mathrm{f} \subseteq D_\mathrm{g}$ $R_f=[0,\infty)\subseteq D_\mathrm{g}=[-\ln 2,\infty)$ Thus gf exists.  Range of gf: $(-3,-2]\xrightarrow{\mathrm{f}}[0,\infty)\xrightarrow{g}[a+1,\infty)$ $R_{\mathrm{gf}}=[a+1,\infty)$	Students can state the condition for gf to exist, but range of f is wrong. For gf, $\mathrm{R_f}=\mathrm{D_g^{new}}$, use the range of f as the new domain of g for composite function.

Hwa Chong Institution (College)

7(i)	Let $u=(\ln x)^2 \quad v'=1$ $u'=2(\ln x)\frac{1}{x} \quad v=x$ $\int (\ln x)^2 \,\mathrm{d}x = x(\ln x)^2 - 2\int \ln x \,\mathrm{d}x$ $\int \ln x \,\mathrm{d}x$ $= x\ln x - \int 1 \,\mathrm{d}x \qquad u=\ln x \quad v'=1$ $= x\ln x - x + C \qquad u'=\frac{1}{x} \quad v=x$ $\int (\ln x)^2 \,\mathrm{d}x = x(\ln x)^2 - 2x\ln x + 2x + C$	This part is well done.
(ii)	When $x=1$, $y=\mathrm{e}^1=\mathrm{e}$ When $x=1$, $y=1+\mathrm{e}-(1)^2=\mathrm{e}$ $\therefore$ (1, e) is the point of intersection of the curves $y=\mathrm{e}^x$ and $y=1-x^2+\mathrm{e}$	Many students compared similar terms and obtained success. For example, these students argued that $1+\mathrm{e}-x^2=\mathrm{e}^x$ implies $\mathrm{e}=\mathrm{e}^x$ and $1=x^2$ which subsequently lead them to conclude that $x=1$. This approach of comparing similar terms in general is not correct. You should convince yourself by looking at the counter example $\mathrm{e}^x+2=\mathrm{e}+x$.
	$y=1-x^2+\mathrm{e}$ $x^2=1+\mathrm{e}-y$ 	There were some students who obtained the same answer using wrong methods as shown:

Hwa Chong Institution (College)

	$V=\pi\int_{e}^{1+e} x^2 dy+\pi\int_{1}^{e}(\ln y)^2 dy$ $=\pi\int_{e}^{1+e} 1+e-y dy+\pi\left[y(\ln y)^2-2y\ln y+2y\right]_1^e$ $=\pi\left[y+ey-\frac{y^2}{2}\right]_e^{1+e}+\pi\left[y(\ln y)^2-2y\ln y+2y\right]_1^e$ $=\pi\left([1+e+e(1+e)-\frac{(1+e)^2}{2}]-[e+e^2-\frac{e^2}{2}]\right)+\pi[e-2e+2e-2]$ $=\pi\left[\frac{1}{2}\right]+\pi[e-2]$ $=\pi\left(e-\frac{3}{2}\right)$	$\pi\int_0^1 1+e-y\ dy-\pi\int_0^1(\ln y)^2\ dy$ $=\pi\left[y+ey-\frac{y^2}{2}\right]_0^1$ $-\pi\left[y(\ln y)^2-2y\ln y+2y\right]_0^1$ $=\pi\left(1+e-\frac{1}{2}-2\right)$ $=\pi\left(e-\frac{3}{2}\right)$. A gentle reminder to this group of students is that the natural logarithm is not defined at 0.
8(i)	$\sin\theta=\dfrac{\begin{pmatrix}0\\1\\1\end{pmatrix}\bullet\begin{pmatrix}1\\2\\-1\end{pmatrix}}{\sqrt{2}\sqrt{6}}=\dfrac{1}{\sqrt{2}\sqrt{6}}$ $\theta=16.8^\circ$ (1 d.pl.)	A few students use $\cos\theta$ and obtain $\theta=73.2^\circ$. They should further obtain $90^\circ-73.2^\circ=16.8^\circ$.
(ii)	Let C be the point of intersection of line l_1 and plane. $\therefore \overrightarrow{OC}=\begin{pmatrix}2\\2+\lambda\\3+\lambda\end{pmatrix}$ for some value of λ. Since C is also on the plane $\prod$, $\begin{pmatrix}2\\2+\lambda\\3+\lambda\end{pmatrix}\bullet\begin{pmatrix}1\\2\\-1\end{pmatrix}=4$ $2+4+2\lambda-3-\lambda=4$ $\lambda=1$ $\overrightarrow{OC}=\begin{pmatrix}2\\3\\4\end{pmatrix}$	Most students scored full marks for (ii) but a few students made careless mstake(s),
(iii)	**Method 1** The line parallel to normal and passing through A is $\mathbf{r}=\begin{pmatrix}2\\2\\3\end{pmatrix}+\alpha\begin{pmatrix}1\\2\\-1\end{pmatrix},\quad \alpha\in\mathbb{R}$.	For **Method 2**, 1) Some students wrongly use $\left\lvert\overrightarrow{AC}\bullet\hat{n}\right\rvert\hat{n}$. The correct formula is $\left(\overrightarrow{AC}\bullet\hat{n}\right)\hat{n}$. 2) Some students got the incorrect

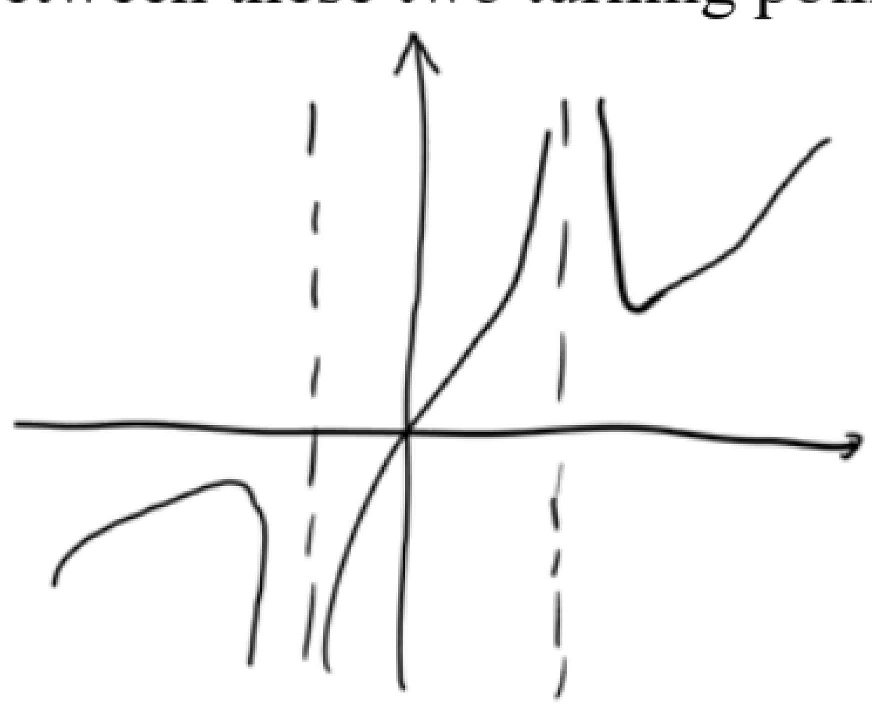
Hwa Chong Institution (College)

	$\begin{pmatrix}2+\alpha\\2+2\alpha\\3-\alpha\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-1\end{pmatrix}=4$ $2+\alpha+4+4\alpha-3+\alpha=4$ $\alpha=\frac{1}{6}$ $\overrightarrow{OF}=\begin{pmatrix}2\\2\\3\end{pmatrix}+\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}=\frac{1}{6}\begin{pmatrix}13\\14\\17\end{pmatrix}$ **Method 2 (Finding projection vector)** $\overrightarrow{AC}=\begin{pmatrix}2\\3\\4\end{pmatrix}-\begin{pmatrix}2\\2\\3\end{pmatrix}=\begin{pmatrix}0\\1\\1\end{pmatrix}$ Let F be foot of perpendicular from A to $\prod$. $\overrightarrow{AF}=\left(\left(\overrightarrow{AC}\cdot\right)\widehat{\begin{pmatrix}1\\2\\-1\end{pmatrix}}\right)\widehat{\begin{pmatrix}1\\2\\-1\end{pmatrix}}$ $=\frac{\left(\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\2\\-1\end{pmatrix}\right)\begin{pmatrix}1\\2\\-1\end{pmatrix}}{\sqrt{6}\sqrt{6}}$ $=\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}$ $\overrightarrow{OF}=\overrightarrow{OA}+\overrightarrow{AF}$ $=\begin{pmatrix}2\\2\\3\end{pmatrix}+\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}=\frac{1}{6}\begin{pmatrix}13\\14\\17\end{pmatrix}$	projected vector $\overrightarrow{FA}=\left(\overrightarrow{AC}\bullet\hat{\mathrm{n}}\right)\hat{\mathrm{n}}$. The correct projected vector is $\overrightarrow{AF}=\left(\overrightarrow{AC}\bullet\hat{\mathrm{n}}\right)\hat{\mathrm{n}}$
(iii) Hence	To find $\overrightarrow{OA'}$: **Method 1 (using parallel vectors)**	A few students wrongly gave the vector equation of the reflected line instead of the Cartesian equation.

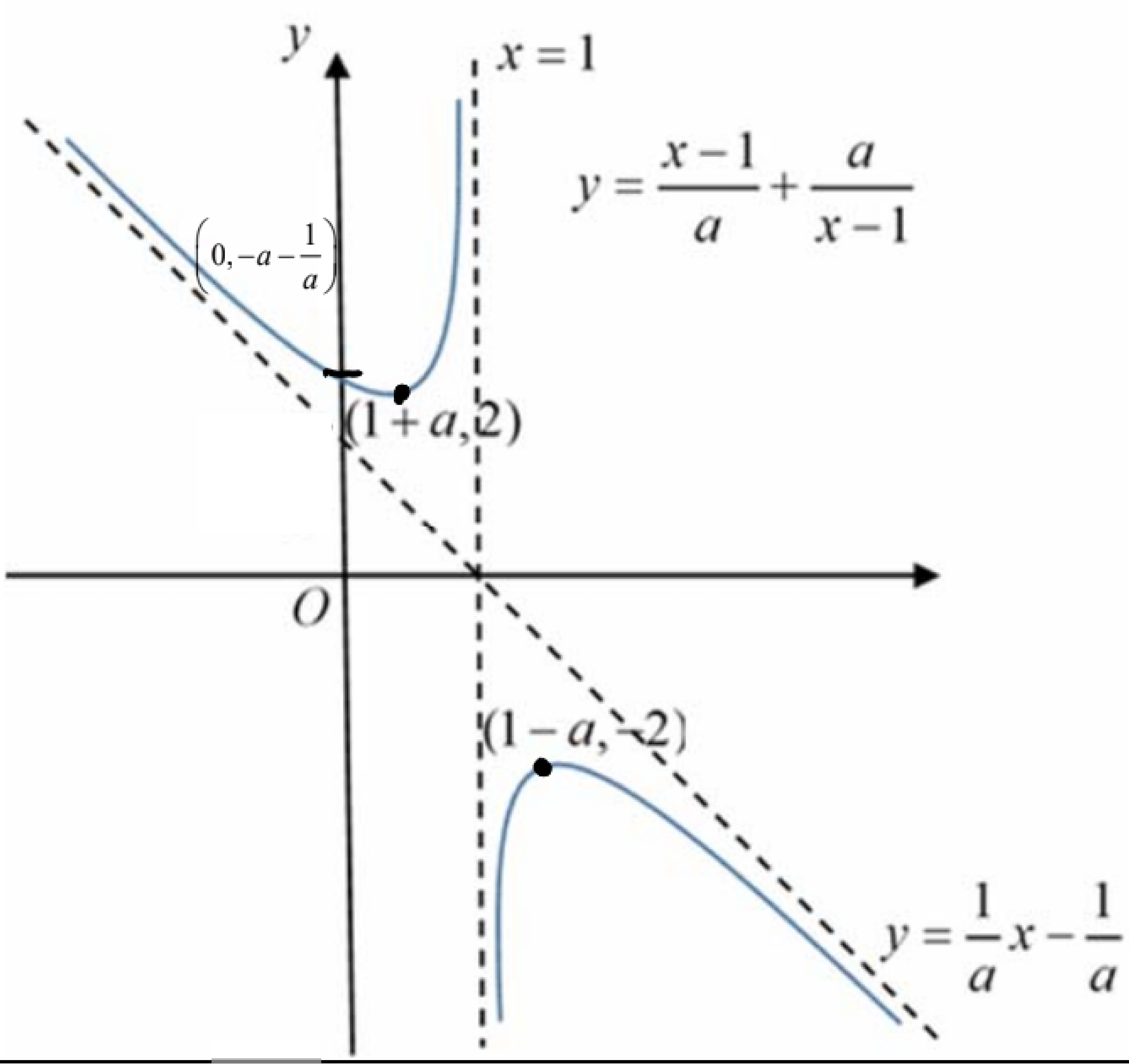
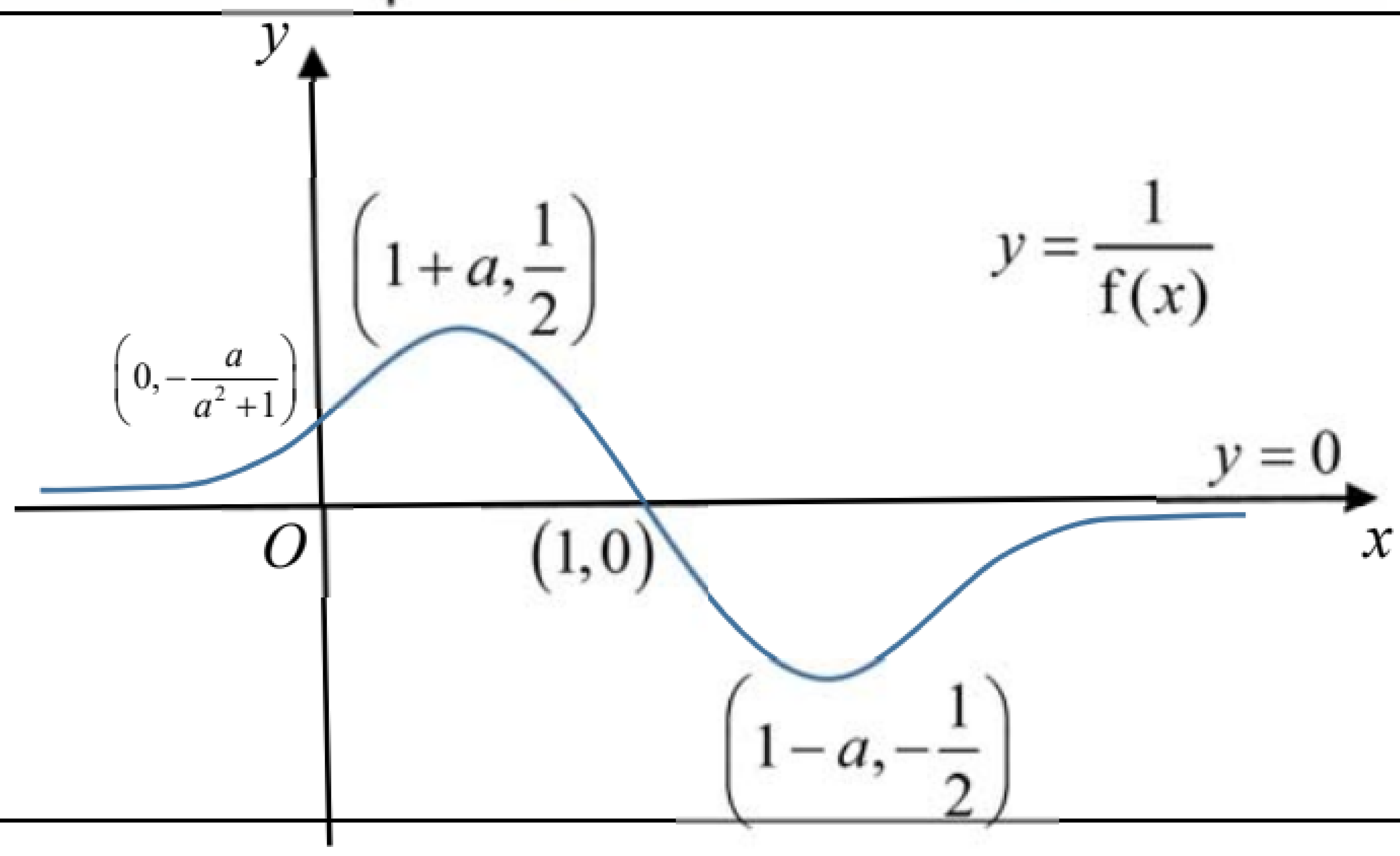
Hwa Chong Institution (College)

	$\overrightarrow{AF}=\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}$ $\overrightarrow{AA'}=2\overrightarrow{AF}=\frac{1}{3}\begin{pmatrix}1\\2\\-1\end{pmatrix}$ $\overrightarrow{OA'}=\overrightarrow{OA}+\overrightarrow{AA'}=\begin{pmatrix}2\\2\\3\end{pmatrix}+\frac{1}{3}\begin{pmatrix}1\\2\\-1\end{pmatrix}=\frac{1}{3}\begin{pmatrix}7\\8\\8\end{pmatrix}$ **Method 2 (using equal vectors)** $\overrightarrow{AF}=\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}$ $\overrightarrow{FA'}=\overrightarrow{AF}=\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}$ $\overrightarrow{OA'}=\overrightarrow{OF}+\overrightarrow{FA'}=\frac{1}{6}\begin{pmatrix}13\\14\\17\end{pmatrix}+\frac{1}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix}=\frac{1}{3}\begin{pmatrix}7\\8\\8\end{pmatrix}$ **Method 3 (using Ratio Theorem)** $\overrightarrow{OF}=\frac{1}{2}\left(\overrightarrow{OA}+\overrightarrow{OA'}\right)$ $\overrightarrow{OA'}=2\overrightarrow{OF}-\overrightarrow{OA}=\frac{1}{3}\begin{pmatrix}13\\14\\17\end{pmatrix}-\begin{pmatrix}2\\2\\3\end{pmatrix}=\frac{1}{3}\begin{pmatrix}7\\8\\8\end{pmatrix}$ Thus $\overrightarrow{A'C}=\overrightarrow{OC}-\overrightarrow{OA'}=\begin{pmatrix}2\\3\\4\end{pmatrix}-\frac{1}{3}\begin{pmatrix}7\\8\\8\end{pmatrix}=\frac{1}{3}\begin{pmatrix}-1\\1\\4\end{pmatrix}$ Thus equation of reflection line is $\mathbf{r}=\begin{pmatrix}2\\3\\4\end{pmatrix}+\beta\begin{pmatrix}-1\\1\\4\end{pmatrix},\quad \beta\in\mathbb{R}.$ Cartesian equation of reflection line is	

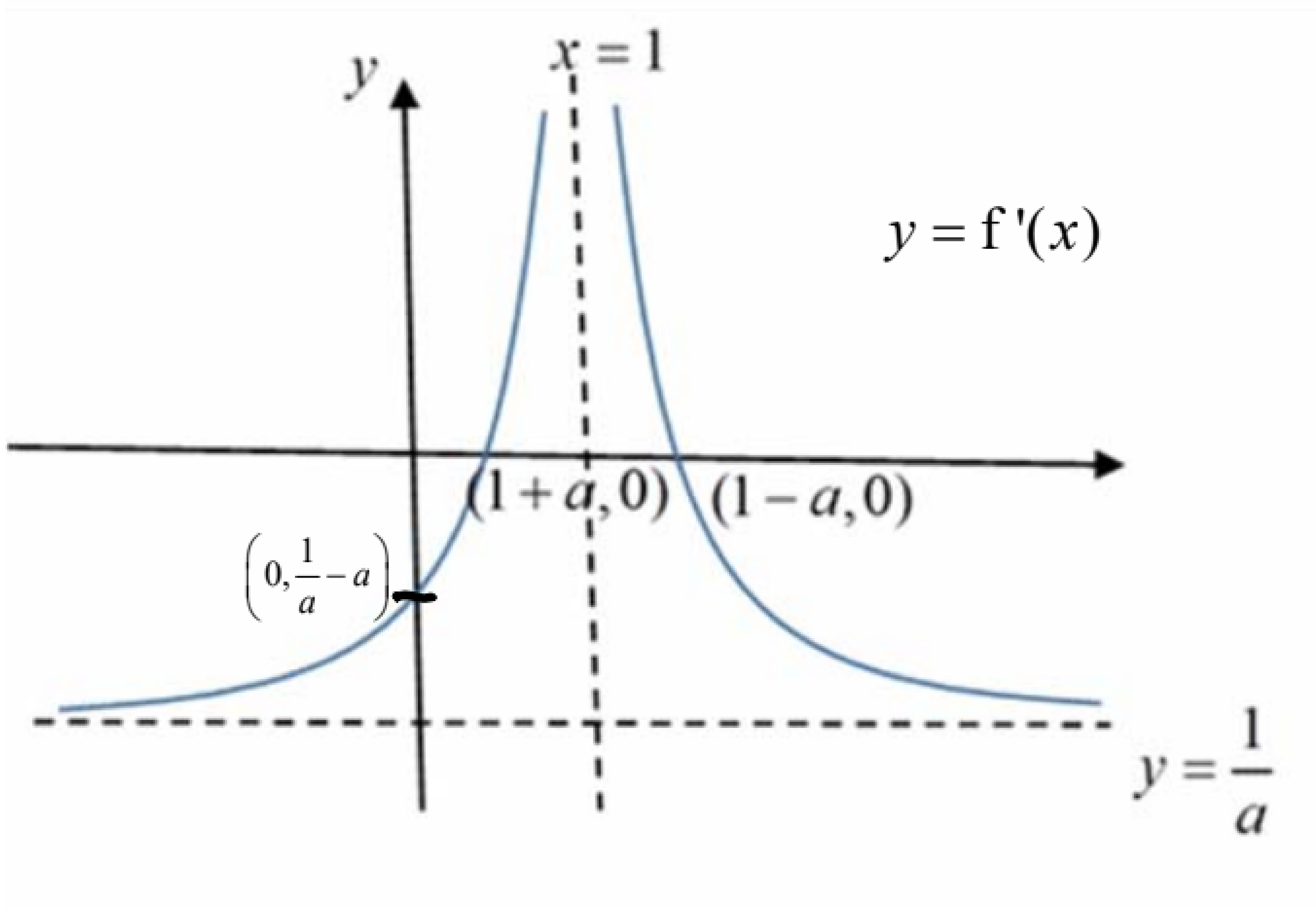
Hwa Chong Institution (College)

	$\dfrac{x-2}{-1}=\dfrac{y-3}{1}=\dfrac{z-4}{4}$.	
(iv)	$l_1:\mathbf{r}=\begin{pmatrix}2\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}0\\1\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$ $l_2:\mathbf{r}=\begin{pmatrix}-1\\4\\7\end{pmatrix}+\mu\begin{pmatrix}-3\\1\\1\end{pmatrix},\quad \mu\in\mathbb{R}$ $\begin{pmatrix}2\\2+\lambda\\3+\lambda\end{pmatrix}=\begin{pmatrix}-1-3\mu\\4+\mu\\7+\mu\end{pmatrix}$ $\Rightarrow\begin{cases}2=-1-3\mu\\2+\lambda=4+\mu\\3+\lambda=7+\mu\end{cases}$ $\Rightarrow$ No solution. Since the lines are non-parallel as $\begin{pmatrix}0\\1\\1\end{pmatrix}\neq k\begin{pmatrix}-3\\1\\1\end{pmatrix}$ for all $k\in\mathbb{R}$ and do not intersect, thus they are skew lines.	A few students did not state that the 2 lines are not parallel.
9(i)	Let $y=\dfrac{x-1}{a}+\dfrac{a}{x-1}$ $\Rightarrow ay(x-1)=(x-1)(x-1)+a^2$ $\Rightarrow x^2-(2+ay)x+1+a^2+ay=0$ Since x is real, discriminant $=(2+ay)^2-4(1)(1+a^2+ay)\geq 0$ $\Rightarrow 4+4ay+a^2y^2-4-4a^2-4ay\geq 0$ $\Rightarrow a^2y^2-4a^2\geq 0$ $\Rightarrow y^2-4\geq 0$ $\Rightarrow (y-2)(y+2)\geq 0$ $\Rightarrow y\leq -2$ or $y\geq 2$ The set of values that $\mathrm{f}(x)$ can take is $\{y: \mathrm{y}\in\mathbb{R}, y\leq -2 \text{ or } y\geq 2\}$	Few students get the correct method for this question. Note that answer must be in **SET** notation. Using turning points is a WRONG method. Below is a graph which has two turning points but it did not imply there is no graph between these two turning points. 
(ii)	$\mathrm{f}(x)=\dfrac{x-1}{a}+\dfrac{a}{x-1}$	In general, things to take note when sketching graphs for (ii) and (iii):

Hwa Chong Institution (College)

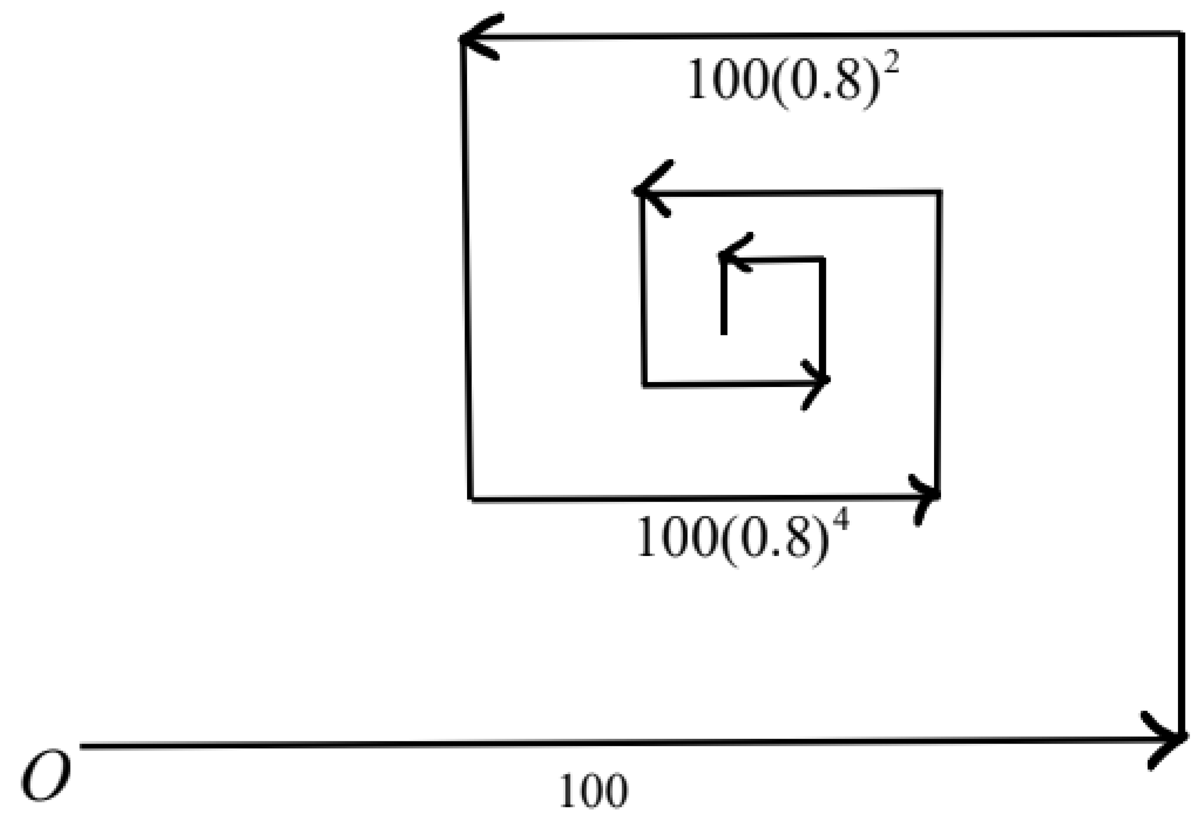
	$f'(x)=\frac{1}{a}+a\left[-(x-1)^{-2}\right]$ At stationary points, $f'(x)=\frac{1}{a}-\frac{a}{(x-1)^2}=0$ $\Rightarrow (x-1)^2=a^2$ $\Rightarrow x=1\pm a$ When $x=1-a, y=-2$ When $x=1+a, y=2$ Equations of asymptotes: $x=1, y=\frac{1}{a}x-\frac{1}{a}$. 	• Sketch STRAIGHT lines with ruler • Sketch a CLEAR curve with key features labelled, use pencil and eraser to adjust • Label asymptote with its equation, not its axial intercepts • Label turning points with its coordinates, do not draw lines to axes to make graph messy • Need to show two or three correct features to get one mark A lot of students did not know how to solve for turning point. This is an ESSENTIAL skill that every student must know. Careless mistakes involved in the differentiation as well. Many students sketch the wrong graph by keying in a positive a value into GC. In this question, a is negative. We only use GC to find out the shape of the graph. For turning points, asymptotes and axial intercepts, need to work out the answers in terms of a.
(iii) (a)		$\frac{1}{-a-\frac{1}{a}}\neq\frac{1}{-a}+\frac{1}{-\frac{1}{a}}$

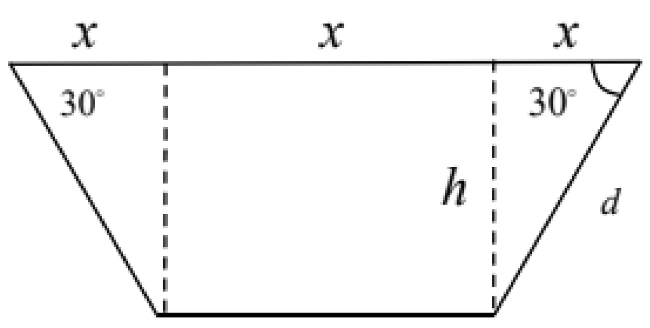
Hwa Chong Institution (College)

<table>
<tr>
<td>(b)</td>
<td>
</td>
<td>The y-intercept is not found by using transformation.
Instead we use $f'(x)=\frac{1}{a}-\frac{a}{(x-1)^2}$ and substitute $x=0$.</td>
</tr>
<tr>
<td>(iv)</td>
<td>$y=\frac{x-1}{a}+\frac{a}{x-1}$
Let $g(x)=\frac{x-1}{a}+\frac{a}{x-1}\xrightarrow[A]{x\to x+1}g(x+1)=\frac{x}{a}+\frac{a}{x}$
$\xrightarrow[B]{y\to 2y}2g(x+1)=\frac{x}{a}+\frac{a}{x}\Rightarrow g(x+1)=\frac{x}{2a}+\frac{a}{2x}$
The curve C is translated in the negative x-direction by 1 unit, followed by scaling parallel to y-axis by a scale factor of $\frac{1}{2}$.
<u>OR</u>
The curve C is scaled parallel to the y-axis by a scale factor of $\frac{1}{2}$, followed by a translation in the negative x-direction by 1 unit.</td>
<td>Incorrect method:
1. Translate in negative x-direction by 1 unit,
$y=\frac{x-1}{a}+\frac{a}{x-1}\to y=\frac{x}{a}+\frac{a}{x}$
2. Scale parallel to x-axis by factor 2
$y=\frac{x}{a}+\frac{a}{x}\to y=\frac{x/2}{a}+\frac{a}{x/2}=\frac{x}{2a}+\frac{2a}{x}$
Notice we get $\frac{2a}{x}$ instead of $\frac{a}{2x}$

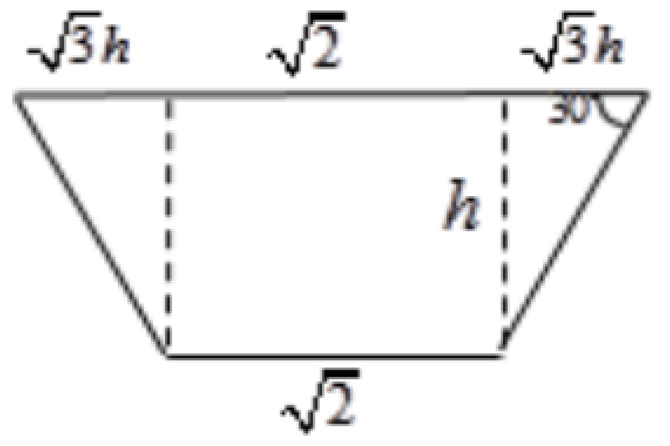
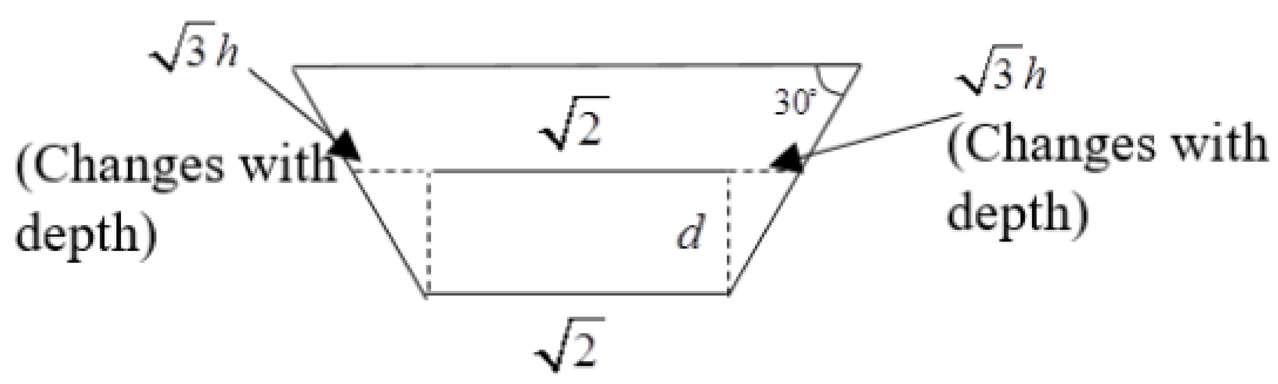
Students must memorize the standard phrasing/words.
‘Translate’, ‘Unit’, ‘Factor’ are key words that cannot be replaced
Advised to use ‘Scale’ instead of ‘Stretch’
Avoid short forms, e.g. write ‘negative’ instead of ‘-ve’. Sometimes handwriting is not clear.</td>
</tr>
<tr>
<td>10 (i)</td>
<td><table>
<tr><td>Between $(n-1)^{\text{th}}$ & n^{th} turn</td><td>Distance covered during the n^{th} turn</td></tr>
<tr><td>1</td><td>100</td></tr>
<tr><td>2</td><td>100(0.8)</td></tr>
<tr><td>3</td><td>$100(0.8)^2$</td></tr>
<tr><td>4</td><td>$100(0.8)^3$</td></tr>
<tr><td>n</td><td>$100(0.8)^{n-1}$</td></tr>
</table></td>
<td>This part of the question is quite well done.</td>
</tr>
</table>

Hwa Chong Institution (College)

	Distance covered between its 11th and 12th turn $= 1000(0.8)^{11} = 8.59$ m	
(ii)	Let the required number of turns be n. $100+100(0.8)+100(0.8)^2+...+100(0.8)^{n-1}=485$ $\dfrac{100\left[1-0.8^n\right]}{1-0.8}=485$ $1-0.8^n=0.97$ $0.8^n=0.03$ $n=\dfrac{\ln 0.03}{\ln 0.8}$ $n=15.7$ The robot has made 15 turns after covering a total distance of 485 m.	Error: (i) $\dfrac{100\left[1-0.8^{n+1}\right]}{1-0.8}=485$ Students must show $n=15.7$. Hence number of turns made is 15.
(iii)	 Eventual horizontal displacement $=100-100(0.8)^2+100(0.8)^4-100(0.8)^6+...$ $=\dfrac{100}{1-\left[-(0.8)^2\right]}$ $=60.976\approx 61.0$ m Eventual vertical displacement $=100(0.8)-100(0.8)^3+100(0.8)^5-...$ $=\dfrac{100(0.8)}{1-\left[-(0.8)^2\right]}\approx 48.8$ m It will end up 61.0 m due east and 48.8 m due north of O. Total distance covered after 8 turns	Students must take note that (iii) is question on sum to infinity and $r=-(0.8)^2$. Students should mention that the robot will end up 61.0m due eat and 48.8m due noth of O. Many students thought total distance covered after 8 turns = $\dfrac{100(1-r^8)}{1-r}$. They are also not certain that $r=1-\dfrac{x}{100}$. Some students cannot obtain the value of r. Some students obtain $x=15.7\%$ instead of 15.7.

	$= 100 + 100\left(1-\frac{x}{100}\right) + 100\left(1-\frac{x}{100}\right)^2 + ... + 100\left(1-\frac{x}{100}\right)^8$ $= \frac{100(1-r^9)}{1-r}$ where $r = 1-\frac{x}{100}$ $\therefore \frac{100(1-r^9)}{1-r} = 500$ $\Rightarrow r^9 - 5r + 4 = 0 ------(1)$ $r = -1.30$ (NA), 1 (NA), 0.84300178 Thus, $1-\frac{x}{100} = 0.84300178$ $\Rightarrow x = 15.7$	
11 (i)	Let h be the height of the trapezium, and d be the diagonal length of trapezium.  $\frac{x}{d} = \cos 30^\circ$ => $d = \frac{2}{\sqrt{3}}x$ We have $\tan 30^\circ = \frac{h}{x}$ $h = \frac{x}{\sqrt{3}}$ Volume of the water trough = $\frac{(3+1)x^2y}{2}\tan 30^\circ$ $= \frac{2x^2y}{\sqrt{3}} = \frac{2}{\sqrt{3}}x^2y$ $\therefore \frac{20}{\sqrt{3}} = \frac{2x^2y}{\sqrt{3}}$ $y = \frac{10}{x^2}$	The expression for the surface area of trough is given, hence all working must be shown clearly. The trough in the question is a open trough, hence there is no need to find the surface area of *ABEF*

Hwa Chong Institution (College)

	$A = xy + 2(d)(y) + 2\left(\dfrac{3x+x}{2}\right)(h)$ $= x\left(\dfrac{10}{x^2}\right) + 2\left(\dfrac{2x}{\sqrt{3}}\right)\left(\dfrac{10}{x^2}\right) + 2\left(2x\left(\dfrac{x}{\sqrt{3}}\right)\right)$ $= \dfrac{10}{x} + \dfrac{40}{x\sqrt{3}} + \dfrac{4x^2}{\sqrt{3}}$ $= \dfrac{4x^2}{\sqrt{3}} + \left(\dfrac{40\sqrt{3}+30}{3x}\right)$ (Shown)	
(ii)	$\dfrac{dA}{dx} = \dfrac{8}{\sqrt{3}}x - \left(\dfrac{40\sqrt{3}+30}{3}\right)x^{-2} = 0$ $x^3 = \dfrac{(40\sqrt{3}+30)\sqrt{3}}{24}$ $x = 1.93\,\text{m}$ $\dfrac{d^2A}{dx^2} = \dfrac{8}{\sqrt{3}} + 2\left(\dfrac{40\sqrt{3}+30}{3}\right)x^{-3} > 0$ when $x = 1.93$ $x = 1.93$ m when A is minimum.	After obtaining the answer for x, there is a need to carry out either the first or second derivative test to show that when $x = 1.93$, the minimum area is obtained. For the test, note that it should be $\dfrac{dA}{dx}$ or $\dfrac{d^2A}{dx^2}$ and not $\dfrac{dy}{dx}$ or $\dfrac{d^2y}{dx^2}$
(iii)	When $y = 5$, $\dfrac{2x^2(5)}{\sqrt{3}} = \dfrac{20}{\sqrt{3}}$ $\Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}$  $\therefore$ Volume of liquid feed V $= \dfrac{(2\sqrt{2}+2\sqrt{3}h)}{2} \times 5 \times h$ $= 5\left(\sqrt{2}+\sqrt{3}h\right)h = 5\sqrt{2}h + 5\sqrt{3}h^2$ $\therefore \dfrac{dV}{dh} = 5\sqrt{2} + 10\sqrt{3}h$	The volume to be found is different from (i). In part one, we find the volume of the trough. But for this question, we should look for the volume of the feed. Referring to the diagram below, the longer of the parallel sides are made up of 3 parts. Two that changes with depth and one that is constant. 

Hwa Chong Institution (College)

$\sqrt{2}$
change with
time)

	When $t = 72$, Volume $= \dfrac{20}{\sqrt{3}} - 0.72$ $\therefore 5\sqrt{2}h + 5\sqrt{3}h^2 = \dfrac{20}{\sqrt{3}} - 0.72$ $\Rightarrow h = 0.7820717$ $\dfrac{\mathrm{d}V}{\mathrm{d}t} = \dfrac{\mathrm{d}V}{\mathrm{d}h} \cdot \dfrac{\mathrm{d}h}{\mathrm{d}t}$ $-0.01 = \left(5\sqrt{2} + 10\sqrt{3}\right)\left(0.7820717\right) \times \dfrac{\mathrm{d}h}{\mathrm{d}t}$ $\dfrac{\mathrm{d}h}{\mathrm{d}t} = -0.000485 \text{ m/hr}$ The rate at which the depth of the water trough is decreasing after 72 hours is 0.000485 m/hr	

Hwa Chong Institution (College)